

PROBLEM STATEMENT

Sum Rule & Power Chain Rule

Find the derivative with respect to x of the function:

$$y = \sin^2 x + \cos^3 x$$

STEP 1 প্রদত্ত ত্রিকোণমিতিক পদদ্বয়কে স্পষ্ট ঘাত আকারে প্রকাশ করি (Expressing Terms in Explicit Power Notation)

$$y = (\sin x)^2 + (\cos x)^3$$

STEP 2 উভয়পক্ষে যোগ বিধি (Sum Rule) অনুসারে অন্তরীকরণ অপারেটর প্রয়োগ করি (Applying Sum Rule for Differentiation)

$$\frac{dy}{dx} = \frac{d}{dx} [(\sin x)^2] + \frac{d}{dx} [(\cos x)^3]$$

STEP 3 প্রথম পদ $(\sin x)^2$ -এর অন্তরজ নির্ণয় করি (Differentiating First Term via Chain Rule)

$$\frac{d}{dx} [(\sin x)^2] = 2 \sin x \cdot \frac{d}{dx} (\sin x) = 2 \sin x \cos x = \sin 2x$$

STEP 4 দ্বিতীয় পদ $(\cos x)^3$ -এর অন্তরজ নির্ণয় করি (Differentiating Second Term via Chain Rule)

$$\frac{d}{dx} [(\cos x)^3] = 3 \cos^2 x \cdot \frac{d}{dx} (\cos x) = 3 \cos^2 x (-\sin x) = -3 \sin x \cos^2 x$$

STEP 5 উভয় অন্তরজ পদ একত্রিত করে উৎপাদকে বিশ্লেষণ করি (Combining Terms & Factoring)

$$\frac{dy}{dx} = 2 \sin x \cos x - 3 \sin x \cos^2 x = \sin 2x - 3 \sin x \cos^2 x = \sin x \cos x (2 - 3 \cos x)$$

FINAL ANSWER $\frac{dy}{dx} = 2 \sin x \cos x - 3 \sin x \cos^2 x$ (or, $\sin 2x - 3 \sin x \cos^2 x$)

BOX 1: MATHEMATICAL FOUNDATIONS & FORMULA REFERENCE

- **Sum Rule:** $\frac{d}{dx}[u + v] = \frac{du}{dx} + \frac{dv}{dx}$.

- **Trigonometric Power Chain Rules:**

$$\frac{d}{dx}[\sin^2 x] = 2 \sin x \cos x = \sin 2x, \quad \frac{d}{dx}[\cos^3 x] = -3 \cos^2 x \sin x$$

- **Common Factored Form:** $\sin x \cos x (2 - 3 \cos x)$.

⚠ BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Negative Sign on Cosine Derivative:** $\frac{d}{dx}(\cos x) = -\sin x$. Forgetting the minus sign produces an incorrect plus sign: $+3 \sin x \cos^2 x$.
- **Exponent Reduction Error:** Forgetting to reduce power of cosine from 3 to 2.
- **Confusion with Multi-Angle Functions:** $\cos^3 x \neq \cos(3x)$. The derivative of $\cos(3x)$ is $-3 \sin(3x)$.

🌐 BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS

- **Domain:** Polynomial trigonometric functions are defined on all real numbers:

$$\text{Domain} = (-\infty, \infty) = \mathbb{R}$$

- **Periodicity:** The function has fundamental period 2π .
- **Smoothness:** Infinitely differentiable (C^∞) everywhere on $\mathbb{R}$.

💡 BOX 4: MULTIPLE-ANGLE IDENTITY TRANSFORMATION METHOD

- Expressing using multiple angle identities:

$$\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^3 x = \frac{\cos 3x + 3 \cos x}{4}$$

$$y = \frac{1}{2} - \frac{1}{2} \cos 2x + \frac{3}{4} \cos x + \frac{1}{4} \cos 3x$$

- Differentiating:

$$\frac{dy}{dx} = \sin 2x - \frac{3}{4} \sin x - \frac{3}{4} \sin 3x$$

- Using $\sin 3x + \sin x = 2 \sin 2x \cos x$, this simplifies exactly to $\sin 2x - 3 \sin x \cos^2 x$.

🎯 BOX 5: EXAM MARKS BREAKDOWN (AMIE SECTION-A FORMAT)

- **Step 1-2 (Sum Rule Setup):** 1.0 Mark
- **Step 3 (First Term Derivative with Chain Rule):** 1.0 Mark
- **Step 4 (Second Term Derivative with Sign Handling):** 1.0 Mark
- **Step 5 & Final Answer:** 1.0 Mark
- **Total Allocation:** 4.0 Marks